

LINEAR ALGEBRA

- Algebra is a branch of mathematics in which arithmetical operations and formal manipulation are applied to abstract symbols rather than specific numbers
- Linear Algebra is a branch of mathematics that lets you concisely describe coordinates and interactions of planes in higher dimensions and perform operations on them
- study of vectors and linear functions
- main building blocks are linear equations, vectors and matrices, linear transformations, determinants and vector spaces

Imp Applications:

- 1) Data set and Data files
- 2) Images and Photographs
- 3) Data Preparation : One hot encoding

The process of converting categorical data into a numerical format that machine learning algorithm can understand

- 4) Linear Regression
- 5) Regularization
- 6) Principal Component Analysis

- 4) Recommendation Systems
- 5) Deep Learning

Vectors :- is a one-dimensional array of numbers

- * dimensionality: the no. of elements in the vector
- * orientation: column orientation or row orientation

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

3D column vector

$$y = [1 \ 2 \ 3]$$

3D row vector

vectorAsList = [1, 2, 3, 4, 5]

vectorAsArray = np.array([1, 2, 3, 4, 5])

rowVector = np.array([[1, 2, 3, 4, 5]])

columnVector = np.array([[1], [2], [3], [4], [5]])

Dot multiplication

$$a \cdot b = b \cdot c$$
$$a \cdot (b + c) = a \cdot b + a \cdot c$$

$$a \cdot b = \sum_{i=1}^n a_i b_i$$

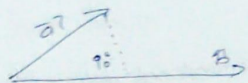
$$\vec{a} = [1, 2, 3, 4, 5]$$

$$\vec{b} = [6, 7, 8, 9, 10]$$

$$a \cdot b = 1 \cdot 6 + 2 \cdot 7 + 3 \cdot 8 + 4 \cdot 9 + 5 \cdot 10$$
$$= 130$$

Magnitude = $\text{np.linalg.norm}(a)$

Vector projection of a vector a onto a vector b



$$\text{formula} = \text{proj}_{b,a} = \frac{a \cdot b}{\|b\|^2} b$$

Vector projection is the process of finding how much of one vector, a , is in the direction of another vector b . You can think of it as the shadow that vector a cast onto a line defined by vector b . The result of a vector projection is another vector that points in the same direction of b , but with a length equal to the shadow.

Basic vectors

→ e_1 and e_2 are called unit vectors along the x-axis and y-axis $(1, 0)$ $(0, 1)$

→ any vector can be represented uniquely as a combination of ^{basic unit} vectors

→ Unit vectors or basic vectors form a basis for space
→ Any vector in space can be written as linear combination of these two vectors.

→ If dot products are zero, then orthogonal

$$e_1 = [1, 0]$$

$$e_2 = [0, 1]$$

$$v = [2, 3]$$

$$= 2[1, 0] + 3[0, 1]$$

new basis

$$b_1 = [1, 1]$$

$$b_2 = [2, 3]$$

$$b_2 = [-1, 1]$$

Find new coordinates of vector v in this new basis

Projection v onto b_1

$$\text{proj}_{b_1,v} = \frac{v \cdot b_1}{\|b_1\|^2}$$

$$v \cdot b_1 = (2)(1) + (3)(1) = 5$$

$$\|b_1\|^2 = 1^2 + 1^2 = 2$$

$$c_1 = 5/2 = 2.5$$

$$\text{proj}_{b_2,v} = \frac{v \cdot b_2}{\|b_2\|^2} =$$

$$v \cdot b_2 = (2)(-1) + (3)(1) = 1$$

$$\|b_2\|^2 = (-1)^2 + 1^2 = 2$$

$$c_2 = 1/2 = 0.5$$

$$v_{\text{new}} = [c_1, c_2] = [2, 1, 0.5]$$

Spanning Set:

Types of matrix transformation:-

Scaling

Inversion

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Rotation

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \text{ } \left. \vphantom{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}} \right\} \text{ rotation matrix}$$